



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

PHYSICS

9188/1

PAPER 1 Multiple Choice

NOVEMBER 2015 SESSION

1 hour

Additional materials:

Electronic calculator and/or Mathematical tables

Multiple Choice answer sheet

Soft clean eraser

Soft pencil (Type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score **one** mark. A mark will **not** be deducted for a wrong answer. Any rough working should be done in this booklet.

This question paper consists of 17 printed pages and 3 blank pages.

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Data

speed of light in free space,	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
permeability of free space,	$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$
permittivity of free space,	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$
elementary charge,	$e = 1.60 \times 10^{-19} \text{ C}$
the Planck constant,	$h = 6.63 \times 10^{-34} \text{ J s}$
unified atomic mass constant,	$u = 1.66 \times 10^{-27} \text{ kg}$
rest mass of electron,	$m_e = 9.11 \times 10^{-31} \text{ kg}$
rest mass of proton,	$m_p = 1.67 \times 10^{-27} \text{ kg}$
molar gas constant,	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
the Avogadro constant,	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
the Boltzmann constant,	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant,	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
acceleration of free fall,	$g = 9.81 \text{ m s}^{-2}$

Formulae

uniformly accelerated motion,

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

work done on/by a gas,

$$W = p\Delta V$$

gravitational potential,

$$\phi = -\frac{Gm}{r}$$

refractive index,

$$n = \frac{1}{\sin C}$$

resistors in series,

$$R = R_1 + R_2 + \dots$$

resistors in parallel,

$$1/R = 1/R_1 + 1/R_2 + \dots$$

electric potential,

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

capacitors in series,

$$1/C = 1/C_1 + 1/C_2 + \dots$$

capacitors in parallel,

$$C = C_1 + C_2 + \dots$$

energy of charged capacitor,

$$W = \frac{1}{2}QV$$

alternating current/voltage,

$$x = x_0 \sin \omega t$$

hydrostatic pressure,

$$p = \rho gh$$

pressure of an ideal gas,

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

radioactive decay,

$$x = x_0 \exp(-\lambda t)$$

decay constant,

$$\lambda = \frac{0.693}{t_{1/2}}$$

critical density of matter in the Universe,

$$\rho_0 = \frac{3H_0^2}{8\pi G}$$

equation of continuity,

$$Av = \text{constant}$$

Bernoulli equation (simplified),

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

Stokes' law,

$$F = 6\pi\eta r v$$

Reynolds' number,

$$R_e = \frac{\rho v r}{\eta}$$

drag force in turbulent flow,

$$F = Br^2 \rho v^2$$

1 What are the base units of the magnetic flux density?

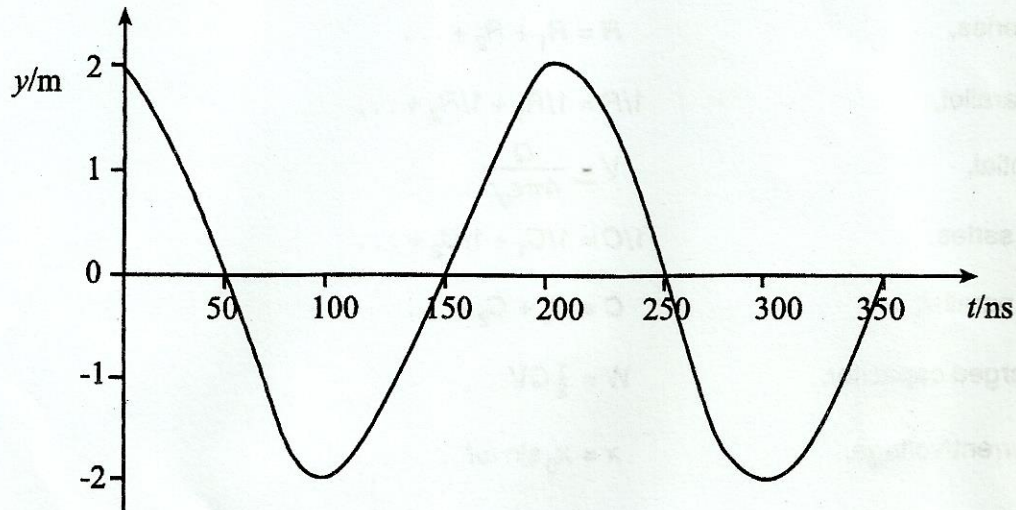
- A $\text{kg s}^{-2} \text{A}^{-1}$
- B $\text{kg m}^{-1} \text{s}^{-2} \text{A}^{-1}$
- C $\text{kg m s}^{-1} \text{A}^{-1}$
- D kg m^{-3}

$$E = BIL$$

$$B = \frac{\text{kg m s}^{-2}}{\text{A m}}$$

$$= \text{kg s}^{-2} \text{A}^{-1}$$

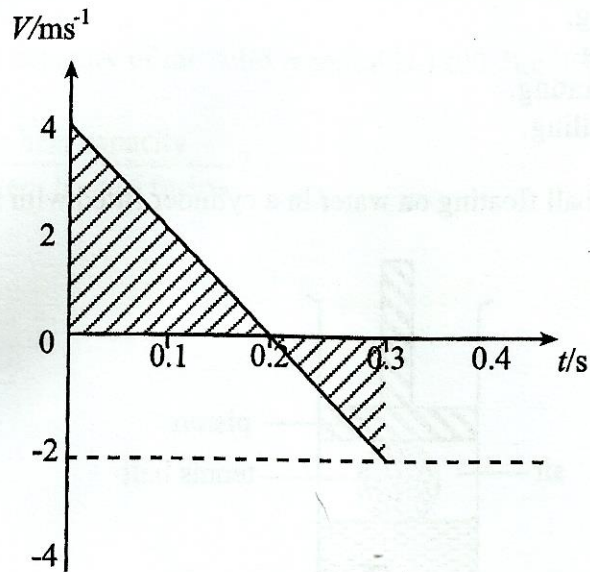
2 The diagram shows a signal displayed on a C.R.O screen.



What is the frequency of the signal?

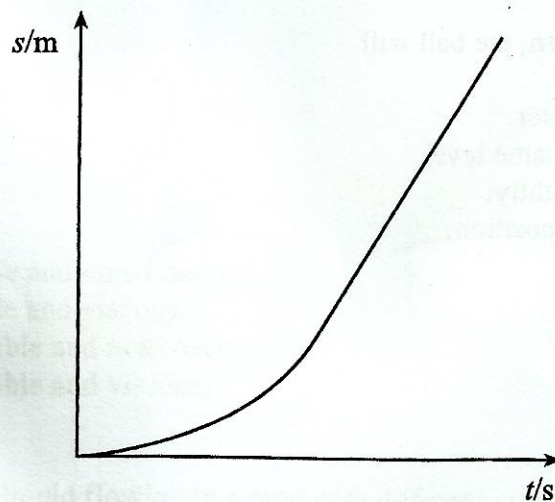
- A 5 mHz
- B 200 Hz
- C 2 MHz
- D 5 MHz

- 3 The velocity-time graph shows the motion of a pellet thrown vertically upwards.



The graph suggests that the

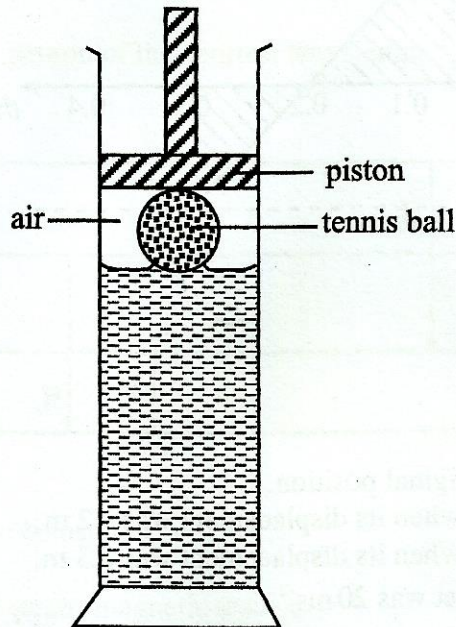
- A pellet returned to its original position.
 - B pellet was intercepted when its displacement was -2 m.
 - C pellet was intercepted when its displacement was 0.3 m.
 - D acceleration of the pellet was 20 ms^{-2} .
- 4 The graph shows the motion of a body released from rest falling through air.



It can be deduced that air resistance is **not** negligible because the velocity

- A always increase with time.
- B eventually increases with time.
- C eventually becomes constant.
- D is always constant.

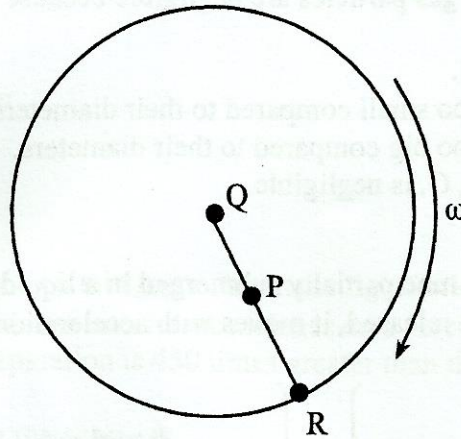
- 5 Which statement does **not** show that there is a resultant force acting on a particle?
- A The particle is moving.
 - B The particle is turning.
 - C The particle is accelerating.
 - D The particle is free falling.
- 6 The diagram shows a tennis ball floating on water in a cylinder fitted with an airtight piston.



When the piston is pressed down, the ball will

- A sink deeper into the water.
- B continue to float at the same level.
- C rise out of the water slightly.
- D vibrate about the same position.

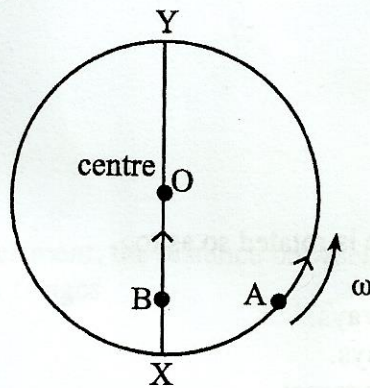
- 7 The diagram shows a circular disc being rotated about its centre Q at an angular velocity ω .



Which relationship between the speed of P, V_P and the speed of R, V_R is correct?

- A $V_P = V_R$
- B $V_P > V_R$
- C $V_P \geq V_R$
- D $V_P < V_R$

- 8 A particle A moves in a circle, centre O. Particle B is a projection of particle A on a screen, XY, placed along the diameter of the circle. As particle A moves around the circle, its projection B moves along XY.



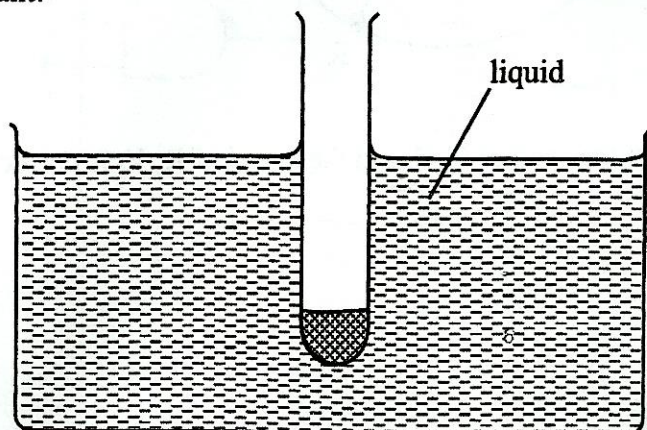
What time is taken by particle B to move from point X to point Y if the period of particle A is 24 hours?

- A 12 hrs
- B 24 hrs
- C 36 hrs
- D 48 hrs

9 The gravitational forces between gas particles are negligible because

- A their masses are too small.
- B their mean separation is too small compared to their diameters.
- C their mean separation is too big compared to their diameters.
- D the gravitational constant, G , is negligible.

10 The diagram shows a loaded test-tube partially submerged in a liquid. When the test-tube is displaced down a distance, x , and released, it moves with acceleration, a , given by $a = -kx$, where k is a constant.



What kind of motion is the test-tube undergoing?

- A undefined sinusoidal
- B simple harmonic
- C longitudinal wave
- D transverse wave

11 In the production of X-rays, the anode is rotated so as to

- A increase the intensity of the x-rays.
- B reduce the intensity of the x-rays.
- C increase the surface area of the target.
- D decrease the surface area of the target.

12 X-rays:

1. are highly penetrating
2. blacken a photographic film
3. are absorbed differently by different materials

Which of the three properties are used in the imaging of internal body structures?

- A 1 and 2
- B 1 and 3
- C 2 and 3
- D 1, 2 and 3

13 Which electromagnetic waves have frequencies in the range 10^{11} to 10^{14} Hz?

- A X-rays
- B infrared radiation
- C gamma rays
- D visible light

EMV D X G
frequency

14 The first order pattern of a certain monochromatic light incident normally on a grating is at 35° . The same light produces interference fringes on a screen 0.90 m from a double slit arrangement. The slit separation is 450 times greater than the line spacing of the grating.

The fringe separation on the screen is

- A 1.1×10^{-3} nm.
- B 1.1×10^{-3} m.
- C 1.1×10^{-3} mm.
- D 1.1×10^{-3} μ m.

$$x = \frac{\lambda D}{d}$$

$$x = \frac{10.90}{450d}$$

$$d = \frac{0.90}{450x}$$

$$d \sin \theta = \lambda n$$

$$\sin 35^\circ = \frac{\lambda}{d}$$

$$d = \frac{\lambda}{\sin 35^\circ}$$

$$0.90 \sin 35^\circ = \lambda$$

$$\lambda = \frac{0.90 \sin 35^\circ}{450}$$

15 A beam of monochromatic light of wavelength 650 nm is incident normally on a diffraction grating which has 600 lines per mm.

What is the angular spacing between the second order fringes?

- A 26°
- B 28°
- C 51°
- D 102°

$$d \sin \theta = n \lambda$$

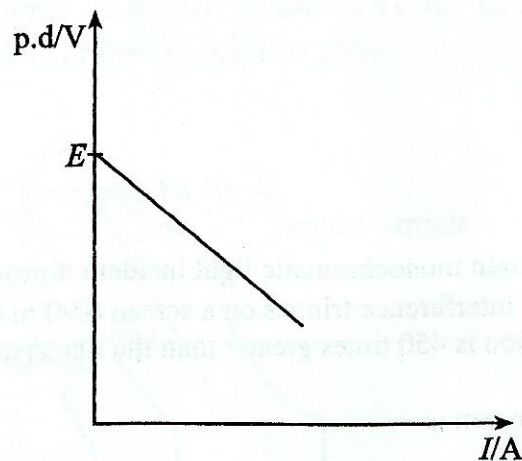
$$\sin \theta = \frac{2(650 \text{ nm})}{600}$$

16 In Young's double slit experiment, the distance between the two slits is halved. The distance between neighbouring dark fringes

- A doubles.
- B halves.
- C increases by a factor of 4.
- D decreases by a factor of 4.

$$x = \frac{\lambda D}{d}$$

- 17 An experiment was carried out to measure the terminal p.d. of a battery of e.m.f., E . The results of the p.d and the current gave the graph shown.



What is the explanation for the decrease in terminal p.d. as current increases?

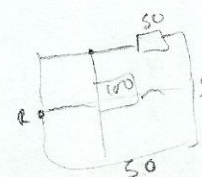
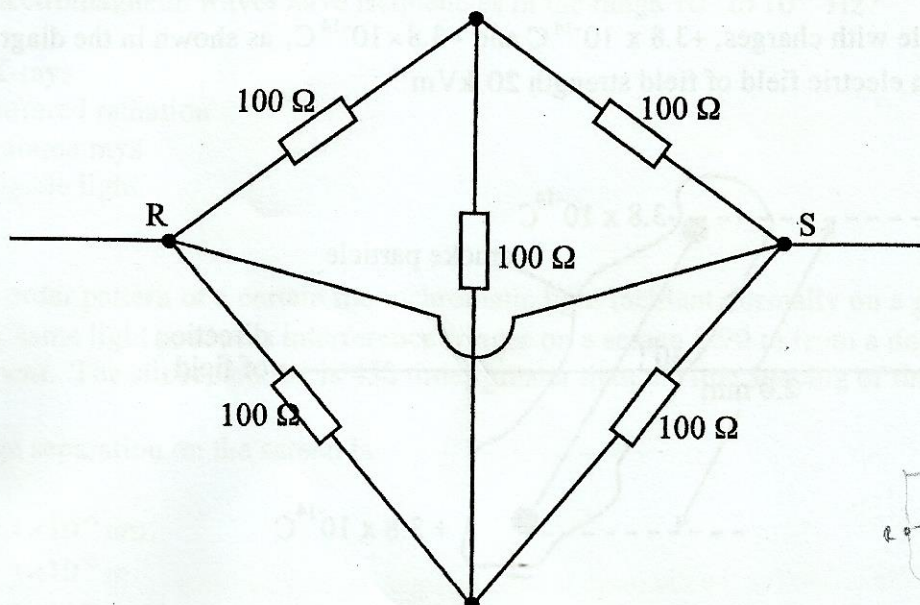
- A internal resistance of the battery increased
- B lost voltage across the internal resistance increased
- C power dissipation across the load is high
- D power dissipation across the internal resistance is low

- 18 A cellphone battery charger gets warm when the battery is charging but cools when the battery is fully charged.

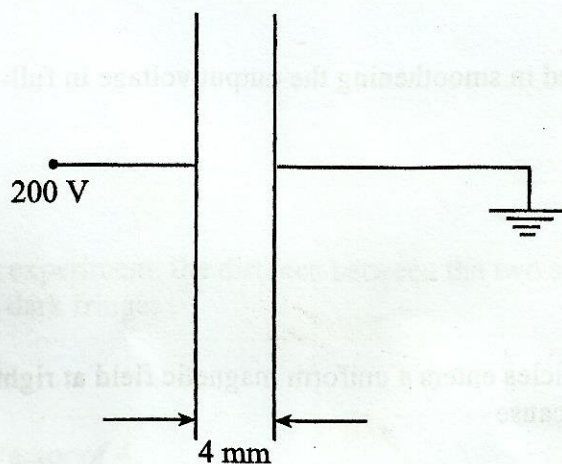
Which row correctly match these changes?

	warm when charging	cool when fully charged
A	there is a net flow of current	there is a net flow of current
B	there is no net flow of current	there is a net flow of current
C	there is a net flow of current	there is no net flow of current
D	there is no net flow of current	there is no net flow of current

- 19 What is the effective resistance between points R and S?



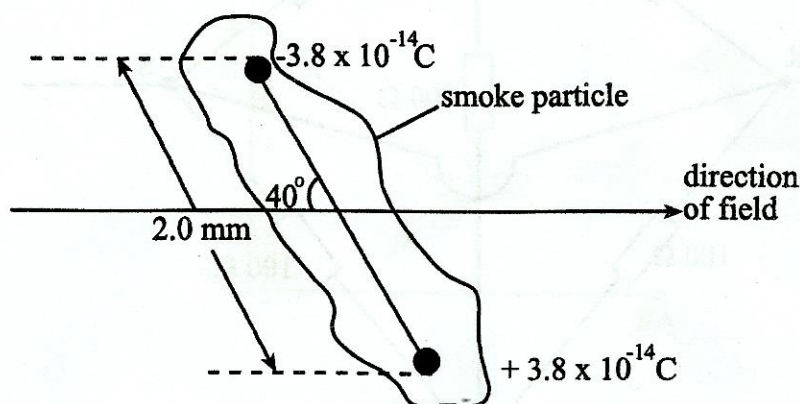
- A 0 Ω
 B 20 Ω
 C 100 Ω
 D 200 Ω
- 20 The diagram shows two parallel plates with one plate earthed and the other one at a potential of 200 V.



The electric field strength is

- A 0 Vm^{-1}
 B 50 Vm^{-1}
 C 10 000 Vm^{-1}
 D 50 000 Vm^{-1}

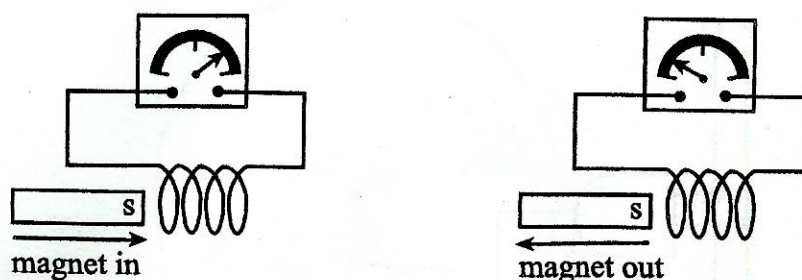
- 21 A smoke particle with charges, $+3.8 \times 10^{-14} \text{ C}$ and $-3.8 \times 10^{-14} \text{ C}$, as shown in the diagram is suspended in an electric field of field strength 20 kVm^{-1} .



What is the magnitude of the couple acting on the particle?

- A $4.6 \times 10^{-13} \text{ Nm}$
 B $9.8 \times 10^{-13} \text{ Nm}$
 C $1.5 \times 10^{-12} \text{ Nm}$
 D $1.9 \times 10^{-12} \text{ Nm}$
- 22 Which electrical device is used in smoothening the output voltage in full-wave rectification?
- A light emitting diode
 B transformer
 C op-amp
 D capacitor
- 23 When a beam of charged particles enters a uniform magnetic field at right angles, in a vacuum, it describes a circular path because
- A there is repulsion between the field and the beam.
 B the beam tries to get out of the field through strong field areas.
 C the force experienced is parallel to the field.
 D the force experienced is at right angles to the direction of motion.

- 24 The diagram shows the deflections of a galvanometer when a magnet is moved into and out of a coil.



Which law is being investigated?

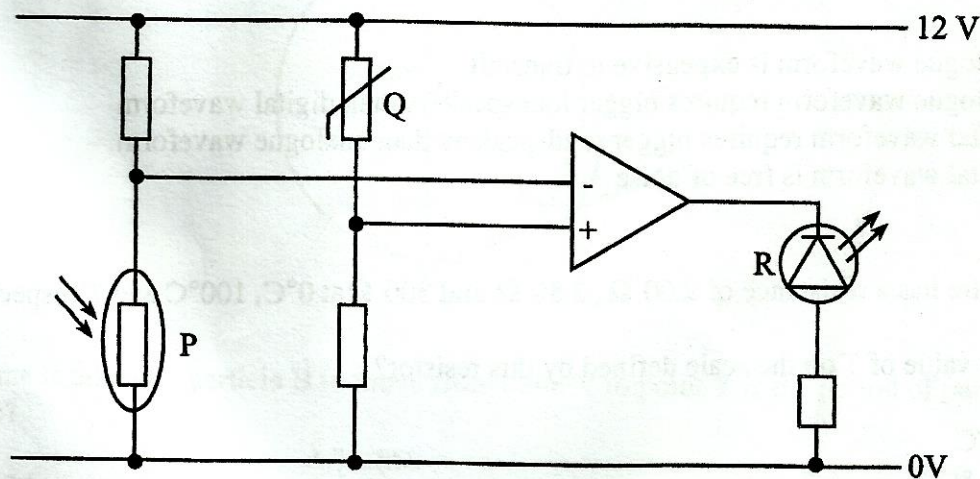
- A Lenz's law
- B Faraday's law
- C Coulomb's law
- D Newton's law

- 25 A sinusoidally varying current has a peak value of 4.0 A.

What is the root-mean-square current?

- A 0.25 A
- B 2.0 A
- C 2.8 A
- D 4.0 A

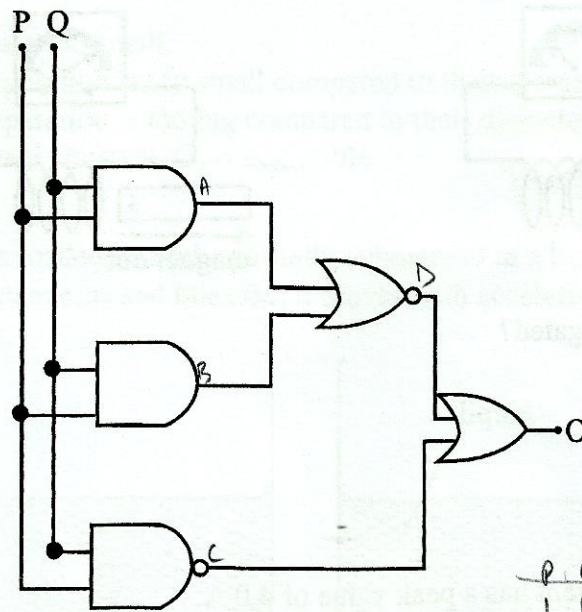
- 26 The diagram shows an op-amp in use.



Which row correctly identifies components P, Q and R?

	P	Q	R
A	Light Dependent Resistor	rheostat	Light Emitting Diode
B	Light Dependent Resistor	thermistor	Light Emitting Diode
C	Light Emitting Diode	rheostat	Light Dependent Resistor
D	Light Emitting Diode	thermistor	Light Dependent Resistor

- 27 What is the equivalent gate for the circuit shown?



P	Q	A	B	C	D	O
1	1	1	1	0	1	1
0	1	0	0	1	0	1
1	0	0	0	1	0	1
0	0	0	0	1	0	1

- A AND
B OR
C NOR
D NAND

- 28 Why is the quality of music transmitted in digital form superior to that transmitted in analogue form?

- A analogue waveform is expensive to transmit
B analogue waveform requires bigger loudspeakers than digital waveform
C digital waveform requires bigger loudspeakers than analogue waveform.
D digital waveform is free of noise

- 29 A coil of wire has a resistance of $2.00\ \Omega$, $2.80\ \Omega$ and $300\ \Omega$ at 0°C , 100°C and T respectively.

What is the value of T on the scale defined by this resistor?

- A 80°C
B 125°C
C 375°C
D $1\ 500^\circ\text{C}$

$$\frac{300 - 2}{2.8 - 2} \times 100$$

- 30 A solid material is heated at a constant rate from 10 °C to a melting point of 90 °C in 100 s. The heating continues for a further 100 s to completely melt the solid material.

If the specific heat capacity of the solid material is $1800 \text{ J kg}^{-1} \text{ K}^{-1}$, what is the ratio

$$\frac{\text{specific heat capacity}}{\text{specific latent heat of fusion}}?$$

- A $\frac{1}{10} \text{ K}^{-1}$
 B $\frac{1}{80} \text{ K}^{-1}$
 C $\frac{1}{90} \text{ K}^{-1}$
 D $\frac{1}{100} \text{ K}^{-1}$

- 31 Air is pumped into a tyre so that the number of moles increases by 12 %, the temperature rises by 11 % and the volume increases by 4 %.

What is the percentage increase of the air pressure in the tyre?

$$PV = nRT$$

$$P = \frac{nRT}{V}$$

- A 19 %
 B 20 %
 C 27 %
 D 33 %

- 32 An ideal fluid is

- A compressible and non-viscous.
 B compressible and viscous.
 C incompressible and non-viscous.
 D incompressible and viscous.

- 33 An incompressible liquid flowing in a pipe with different cross sectional areas reduces speed when it enters a wider part.

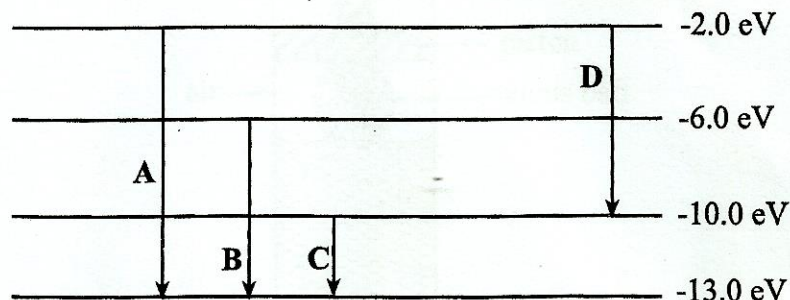
This is in line with the principle of conservation of

- A charge.
 B energy.
 C mass.
 D momentum.

34 If two bodies are in thermal equilibrium, they

- A have zero thermal transfer.
- B have equal amounts of thermal energy.
- C are at the same temperature.
- D always have a net temperature of zero.

35 Which transition would emit a photon of the shortest wavelength?



36 The line emission spectrum can be used to

- A study characteristics of electromagnetic waves.
- B show the continuous spectrum.
- C show quantization of light.
- D identify elements producing them.

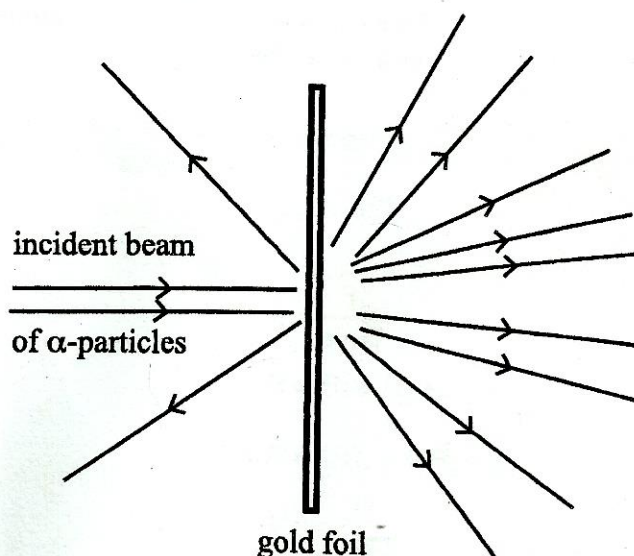
37 The maximum kinetic energy of photoelectrons from a metal surface is 2.6 MeV and the work function of the metal is 0.37 MeV.

What is the wavelength of the incident wave?

- A 6.7×10^{-26} m
- B 4.2×10^{-13} m
- C 5.6×10^{-13} m
- D 7.2×10^{20} m

$$\begin{aligned}
 h\nu &= 2.6 \text{ MeV} \\
 h\nu &= 0.37 \text{ MeV} \\
 \frac{hc}{\lambda} &= 2.6 \text{ MeV} \\
 \frac{hc}{\lambda} &= 0.37 \text{ MeV}
 \end{aligned}$$

- 38 The diagram shows the scattering of alpha particles by a gold foil.



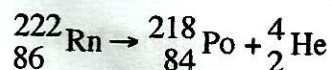
These observations provide evidence that

- A a beam of alpha particles can be diffracted by a gold foil.
 - B all the mass of the atom is concentrated in the nucleus.
 - C much of the atom is empty space.
 - D the beam of alpha particles can be refracted by a gold foil.
- 39 The volume of a uranium-238 nucleus is $2.9 \times 10^{-42} \text{ m}^3$ and that of uranium atom is $5.1 \times 10^{-7} \text{ m}^3$.

If the density of the uranium -238 is $1.95 \times 10^4 \text{ kg m}^{-3}$, then the

- A atom is less massive.
- B volume of the atom is negligible.
- C most part of the atom is empty space.
- D volume of the atom is 1.80×10^{35} times smaller than the volume of nucleus.

- 40 In the radioactive decay,



the mass defect is 0.006 U.

What fraction of the total kinetic energy is the kinetic energy of the α - particle if the radon nucleus is initially stationary?

- A 0.98
- B 0.50
- C 0.20
- D 0.02

Fact 998
J. Marcano p824 (A)